# Group Assignment:

## Build an Action-Oriented Agentic Tutor with SWE-Agent

### The Task

Use SWE-Agent, or any combination of AI coding tools, to build an Action-Oriented Tutor that analyzes a student's learning profile and nudges them toward active thinking and practice.

Inspired by the "learning by doing" philosophy and DeepTutor's RAG-based learner indexing (DeepTutor, https://github.com/HKUDS/DeepTutor). The Action-Oriented Tutor does not lecture or provide direct answers. It identifies where understanding is shallow or stale and provokes the student to do something about it.

***What the Tutor Does***

Input: a student's learning trace (notes, exercise logs, error history, timestamps)

Output: a targeted nudge that prompts thinking or practice; or guided interaction to help yield those nudges.

***Required Architecture***

At least three components should be created, mainly by vibe coding or interaction with SWE-Agent:

[1] Trace Indexer: parses and indexes the student's learning history

[2] Gap Detector: identifies weak, avoided, or decaying concepts from the trace

[3] Nudge Engine: generates a direct, actionable provocation, no explanations, no solutions

### Analytics and Reflection

[1] Analyze the capabilities and the limitations of SWE-Agent that you observed when building the tutor with that.

[2] What have you or the SWE-Agent tried that didn't work?

[3] What surprised you?

[4] The feasibility of turning this prototype into a real application.

[5] Any observation deserves to be shared.

### Deliverables

[1] Working or failed prototype

[2] Prompt log

[3] Report (≤ 4 pages): introduction to the SWE-Agent employed, design rationale, evaluation, capability & limits, AI usage disclosure

[4] Demo (3 min in a video)

## *Note*

*You are allowed to fail in constructing a working prototype, then summarize that in your submission.*

## **Please submit to Canvas before the next lecture.**